

CS & IT ENGINEERING



Algorithms

Greedy Algorithms

Lecture No.- 02



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

→ PYQs on Masters Mtd

→ Intro to Greedy Algo

→ Fractional Knapsack

Topics to be Covered



Topic

Topic

Topic

Fractional Knapsack

Imp
Terminologies



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe

#Q. Find max/optimal solution to the fractional knapsack problem:

Given: $n = 7$, $M = 15$

Object	Profit	Weight
01	10	2
02	5	3
03	15	5
04	7	7
05	6	1
06	18	4
07	3	1

$$\begin{aligned}
 m &= 15 - 1 \\
 &= 14 - 2 \\
 &= 12 - 4 \\
 &= 8 - 5 \\
 &= 3 - 1
 \end{aligned}$$

$$\begin{aligned}
 2 &< 3 \\
 &\underline{\underline{2}} \\
 &= 0
 \end{aligned}$$

P/W

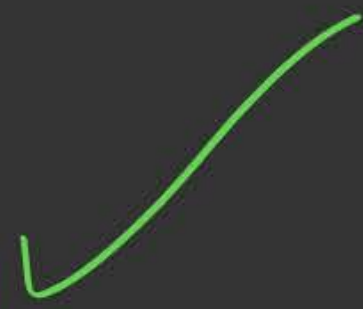
$$\begin{aligned}
 &\rightarrow 10/2 = 5 \text{ --- } (2) \\
 &\rightarrow 5/3 = 1.66 \text{ --- } (6) \\
 &\rightarrow 15/5 = 3 \text{ --- } (4) \\
 &\rightarrow 7/7 = 1 \text{ --- } (7) \\
 &\rightarrow 6/1 = 6 \text{ --- } (1) \\
 &\rightarrow 18/4 = 4.5 \text{ --- } (3) \\
 &\rightarrow 3/1 = 3 \text{ --- } (5)
 \end{aligned}$$

$$\text{max Profit} = \sum P_i X_i$$

$$= 10 + \frac{2}{3} \times 5 + 15 + 6 + 18 + 3$$

$$= 52 + \frac{10}{3}$$

$$= \underline{\underline{55.33}}$$



$$P_{\text{profit}} = \underline{47}$$

$$Wt = \underline{54}$$



Topic : Greedy Algorithm



Procedure GREDY_KNAPSACK(P, W, M, X, n)
// P(1 : n) and W(1 : n) contain the profits and weights respectively of the n//
// knapsack size and X(1 : n) is the solution vector//
Real P(1 : m), W(1 : n), X (1 : n), M, cu;
integer l, n;
x $\leftarrow$ 0 // initialize solution to zero//
cu $\leftarrow$ M // cu = remaining knapsack capacity//
for i $\leftarrow$ 1 to n do
 if W(i) > cu then ~~exit~~ ^{break} endif
 X(i) $\leftarrow$ 1

Assuming:

Objects are sorted by
P/w values.

$$X_i = \frac{cu}{w_i}$$



Topic : Greedy Algorithm



$cu \leftarrow cu - W(i)$

Repeat

if $i \leq n$ then $X(i) \leftarrow (cu/W(i))$ endif

end GREEDY-KNAPSACK

#Q. Consider the weights and values of items listed below. Note that there is only one unit of each item. (by default)

item number	Weight (in kg)	Value (in Rupees)
1	$X_1 = 0$ 10	60
2	$X_2 = 0$ 7	28
3	$X_3 = 1$ 4	20
<u>4</u>	$X_4 = 1$ 2	24

Profits

P/w

$$\begin{aligned}
 60/10 &= 6 - \textcircled{2} \\
 28/7 &= 4 - \textcircled{4} \\
 20/4 &= 5 - \textcircled{3} \\
 24/2 &= 12 - \textcircled{1}
 \end{aligned}$$

$m = 11$

$\{1\}$ $\{2,3\}$ $\{3,4\}$ $\{2,4\}$ $\rightarrow 52$
 60 48 44

$m = 11 - 2 - 4$
 $= 5$

#Q. The task is to pick a subset of these items such that their total weight is no more than 11 kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt} . A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy} . The value of $V_{\text{opt}} - V_{\text{greedy}}$ is ____.

$n=11$

0/1 Knapsack

$$\underline{V_{opt} = ?}$$

$$V_{opt} = 60$$

$$\begin{aligned} V_{\text{greedy}} &= 24 + 20 \\ &= \underline{\underline{44}} \end{aligned}$$

$$\text{Ans} = V_{\text{opt}} - V_{\text{greedy}} = (60 - 44) = \boxed{16} \checkmark$$

Imp Terminologies + Concepts



Topic : Greedy Algorithm



Important Terminologies:

1) Problem Definition: ✓

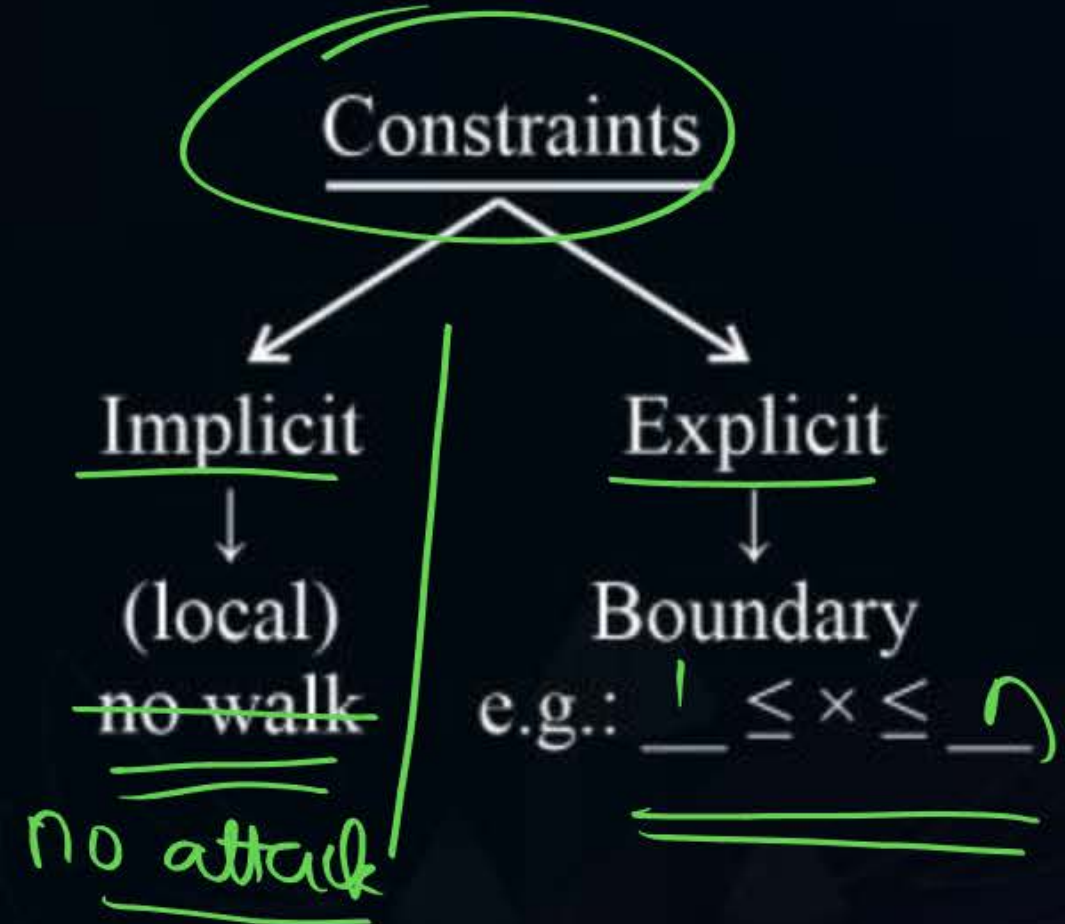
2) Solution Space:

- All possible arrangements
- Only satisfies explicit constraints.

3) ~~Problem Definition:~~ Feasible Soln

- Those solutions from solution space that also satisfies implicit constraints.

Bonus





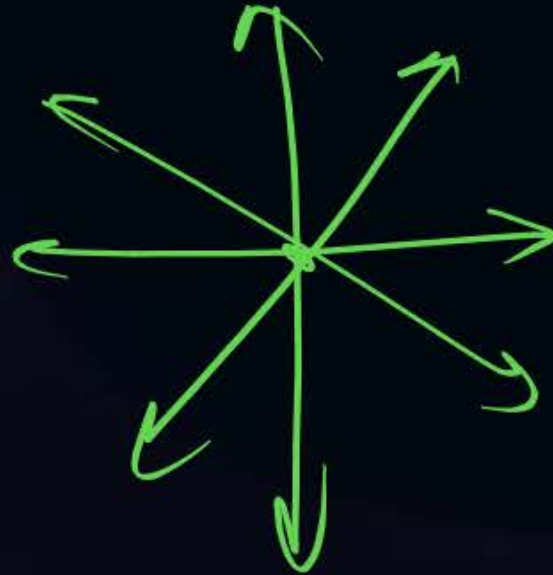
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Example:

N-Queens Problem:

- Given n-Queens on a $n \times n$ board, arrange them such that no two queens attack each other.

E.g.: $n = 4$: $q_1 \quad q_2 \quad q_3 \quad q_4$



	1	2	3	4
q_1				
q_2				
q_3				
q_4				



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E.g.: $n = 4$: q_1 q_2 q_3 q_4

$X[1 \dots n]$

$X[i] \rightarrow$ position of q_i in the i^{th} row

$\{1 \leq X_i \leq n\}$

	1	2	3	4
q_1				
q_2				
q_3				
q_4				

$1 \leq X_i \leq n$

— — — —



Topic : Greedy Algorithm



E.g.: $n = 4$

$X = [1, 2, 3, 4]$

$X = [2, 3, 1, 4]$

$X = [4, 3, 2, 1]$

:

:

:

Size of solution space = $n!$

Solution Space

$$n \times (n-1) \times (n-2) \times \dots \times 1$$

$$= n!$$

$$\underline{\underline{n!}}$$

	1	2	3	4
q_1				
q_2				
q_3				
q_4				



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$$X = [1, 3, _, _]$$

$[1, _, _, _]$ ✗

- This is in solution space,
- But not feasible solution.

→ $X = [1, 2, 3, 4]$

↓
not feasible

	1	2	3	4
q_1	q_1			
q_2		q_2		
q_3			q_3	
q_4				q_4

	1	2	3	4
q_1	q_1			
q_2			q_2	
q_3				
q_4				



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$$X = [2, 4, 1, 3]$$

	1	2	3	4
q_1		q_1		
q_2				q_2
q_3	q_3			
q_4			q_4	

- Both are feasible solution.

mirror

$$X = [3, 1, 4, 2]$$

	1	2	3	4
q_1			q_1	
q_2	q_2			
q_3				q_3
q_4		q_4		

$q_1 = q_4$ (feasible)



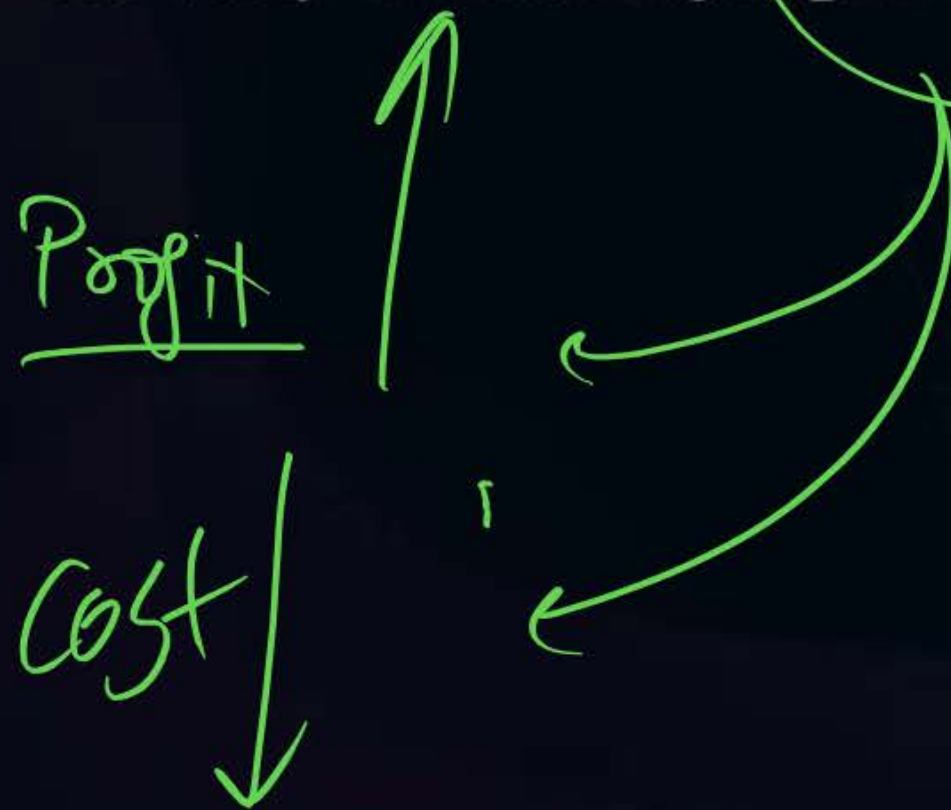
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Important Terminologies:

4) Objective function:

- Optional ✓
- Tries to optimize (minimize/maximize) a given criteria of the problem.





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Example:

- 1) Knapsack → Maximize profit
- 2) Shortest Path → Minimize path lost



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Important Terminologies:

5) Optimal Solution:

- It is that feasible solution that satisfies the objective function.
- Optimal solution always refers to the value and hence it is unique.

↓
 V_{imp}

↓
 R



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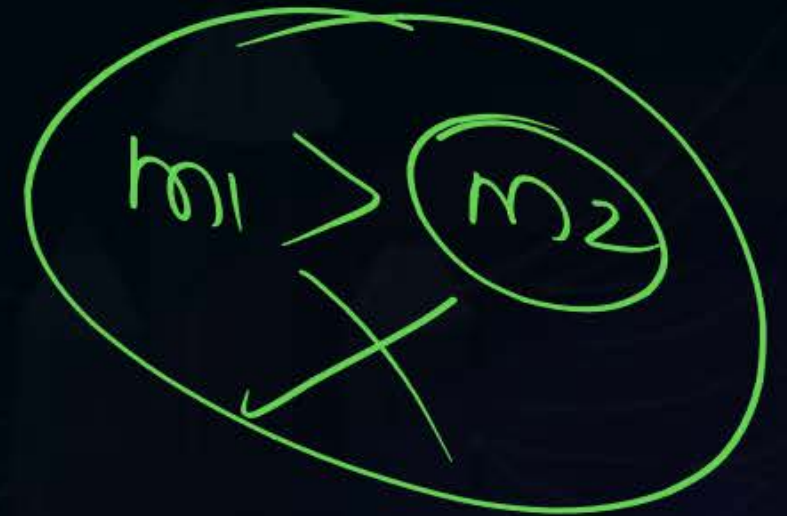
Example:



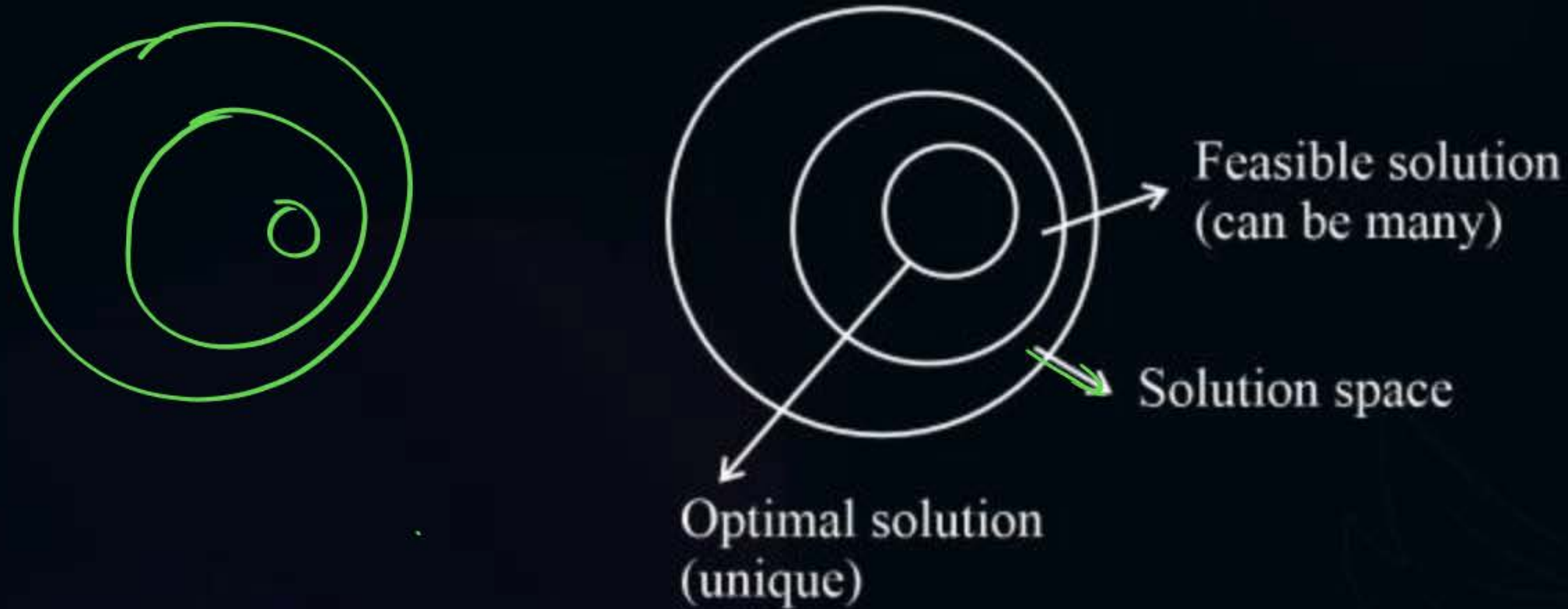
- Min Cost $A \rightarrow C$: 12 (unique)
- Shortest Path $A \rightarrow C$:

$$A \rightarrow B \rightarrow C : 5 + 7 = 12$$

$$A \rightarrow D \rightarrow C : 8 + 4 = 12$$



#Q. What is the objective function of n-Queens problem?



- It does not have any objective function. Hence no optimal solution.
- It does have multiple feasible solutions.



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Problem (P)

Decision Problem

- result is always either True/False.
- Feasible Solution(s)
e.g.: Searching, Sorting, n-Queens etc.

Optimization Problem

- Requires to determine a min/max value (optimal sol).
- Optimal solution
- Objective function
e.g.: Knapsack
Shortest Path Problem



2 mins Summary



Topic

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Fractional Knapsack

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Terminologies

Topic



THANK - YOU